

PRODUCT REQUIREMENTS DOCUMENT

Document Version	1.0
Product Name	Liveness Detection
Document Owner	Alfizdiana Sholikhah
Developer	
Stakeholder	Denis Setiawan
Doc Stage	Ongoing
Created Date	2024-09-02

Overview

The Liveness Detection project aims to develop a system that can accurately distinguish between live images of a user's face and digital representations, such as 2D prints or video clips. This technology is crucial for enhancing security and ensuring the authenticity of facial recognition processes. The solution is designed for deployment in mobile applications, utilizing MTCNN or DeepFace models. It is targeted for five markets: Indonesia, India, the Philippines, Vietnam, and Thailand, providing a localized and robust solution for identity verification across these regions.

Problem Statement

In the rapidly growing digital world, the risk of identity fraud through facial recognition systems is significant. Current solutions often fail to differentiate between a live image and a fake representation, leading to potential security breaches. Key Issue:

Inadequate Differentiation: Many existing facial recognition systems are unable to reliably distinguish between a live person and a spoofed image or video.

Market-Specific Challenges: Different regions require tailored solutions due to varying environmental conditions, facial features, and cultural nuances.

Deployment Complexity: Integrating advanced liveness detection into mobile applications is technically challenging, requiring a lightweight and efficient model that can run smoothly on various devices.

Objective

The objective of the Liveness Detection project is to create a robust and scalable system that accurately verifies the liveness of a user's face, preventing identity fraud and enhancing security in digital identity verification processes.

Key Objective:

Liveness Verification: Develop a model using MTCNN or DeepFace to accurately detect live faces and differentiate them from 2D images or video clips.

Mobile Deployment: Implement the system using Flask or FastAPI for seamless integration into mobile applications, specifically targeting Android (Kotlin) platforms.

Market Localization: Tailor the solution to meet the specific needs of five key markets: Indonesia, India, the Philippines, Vietnam, and Thailand, ensuring high accuracy across diverse user bases.

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	Tags	Guidelines
Decider	Denis Setyawan	As the decider, Eric is responsible for making critical decisions regarding model selection (MTCNN/DeepFace), market-specific adjustments, and the overall system architecture for Liveness Detection.
Accountable		Accountable for the successful implementation and deployment of the Liveness Detection system, ensuring that all key objectives are met within the specified timeline and budget.
Responsible		Responsible for specific tasks related to the development and deployment of the Liveness Detection system, such as model training, Flask/FastAPI API development, mobile application integration (Kotlin), and localization for the targeted markets.
Consulted		Stakeholders are consulted throughout the project to gather feedback on system requirements, user preferences, and desired features. Their input helps ensure that the Liveness Detection system meets the needs and expectations of its intended users.
Informed		Regular updates and progress reports are provided to stakeholders and decision-makers, keeping them informed about the project's status, challenges, and successes. This ensures transparency and alignment with project goals throughout the development process.

Project Timeline

Time Period	Activity	PIC
03 Sep 2024	Grooming	PM
09 - 20 Sep 2024	Model Development (MTCNN/DeepFace)	Team Engineering
23 Sep - 07 Oct 2024	Flask/FastAPI Development	Team Engineering
07 - 18 Oct 2024	Mobile Integration (Kotlin)	Team Engineering
21 Okt - 01 Nov 2024	Testing and Optimization	Team Engineering

Success Metrics

Metrics	Definition	Actual	Target
Liveness Detection Accuracy	% of correctly identified live faces vs. spoofed attempts		95%
API Reliability	Reliability of the API in handling requests and processing data		98%
User Satisfaction	User satisfaction with ease of use and accuracy		90%

User Stories

Title	User Story	Acceptance Criteria	Priority
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Title	User Story	Acceptance Criteria	Priority
Effortless Identity Verification	As a user, I want the system to quickly and accurately verify my identity by detecting live facial movements, so I can securely access my accounts.	Given I use the app for identity verification, when I present my face, the system should correctly verify it as live.	High
Seamless Mobile Integration	As a developer, I want the liveness detection system to be easily integrated into mobile applications, so I can deploy it efficiently on various devices.	Given I integrate the API with a mobile app, the integration should be smooth and not affect app performance.	High
Real-time Feedback	As a user, I want to receive immediate feedback on the liveness detection result, so I can quickly know if my verification is successful or not.	Given I complete a liveness check, the system should instantly notify me of the result.	High

Title	User Story	Acceptance Criteria	Priority
Localization for Accurate Detection	As a user in Indonesia, India, the Philippines, Vietnam, or Thailand, I want the system to accurately detect my face's liveness, considering regional factors.	Given the system is deployed in these regions, it should maintain high accuracy despite diverse facial features and conditions.	Medium